

Kayla McFarlane

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EDUCATION

Carnegie Mellon University

Master of Science in Robotics

Bachelor of Science in Electrical and Computer Engineering (*Software Systems Concentration*)

Pittsburgh, PA

August 2027

May 2025

AWARDS & HONORS

- **Rales Fellow, Carnegie Mellon University (2025–2027)** — Full-tuition fellowship, **34 of 400+** applicants
- **ETH Zurich Robotics Summer School (June 2025)** — Selected as **1 of 50** students globally from **292** applicants

RESEARCH EXPERIENCE

Graduate Research Assistant, CMU AirLab | Safe Agile Foresight Everywhere Project

May 2025 — Present

- Leading development of benchmark to evaluate adaptive safety behavior in autonomous navigation under uncertainty to distinguish genuine risk reasoning from pattern memorization
- Conducted systematic literature review of **20+ papers** on risk-aware navigation, benchmarking, and decision-making.
- Building Python-based testing framework in Isaac Sim with automated stochastic scenario generation
- Authored workshop paper on multi-modal failure adaptation, accepted to IROS 2025 Workshop
- Authoring research paper targeting IROS 2026 conference submission

Undergraduate Research Assistant, CMU AirLab | Wildfire Project

Jan 2023 — May 2025

- Contributed to Wildfire Project, aimed at providing real-time situational awareness to firefighters during wildfires.
- Built automated data collection pipeline generating **15,000+** multimodal training images (RGB, thermal, depth)
- Created comprehensive testing framework to validate pipeline components, ensuring data quality for ML training
- Developed realistic smoke simulation in FIREVision using Unreal Engine for testing drone perception
- Presented research at CMU Meeting of the Minds Symposium and developed demo materials for Epic Games MegaGrant application

Study Bearbot – Multisensory HRI System | Capstone Project

Jan 2025 — May 2025

- Designed and evaluated multisensory desktop robot for student focus and stress reduction
- Conducted **8-participant** user study demonstrating measurable stress reduction and increased focus
- Developed embedded firmware for Arduino Nano RP2040 with real-time sensor integration achieving **sub-500ms** responsiveness
- Built React-based interface for real-time robot control and study session tracking

Human-Robot Interaction Research Project

Nov 2023 – Dec 2023

Exploring Accent Variability, Phrase Difficulty, and Speaker Distance in Robot-Enabled Speech Recognition

- Evaluated Google Speech Recognition integrated with Pepper robot across diverse accents (American, Chinese, Guyanese, Brazilian) to assess ASR performance in classroom settings
- Developed Python modules for real-time transcription and analyzed performance using Translation Error Rate (TER) metrics
- Identified critical limitations in ASR accuracy for accented speech and distance variability, providing recommendations for improving accessibility in educational settings
- Project code adopted by AI Maker Space for demonstrating speech recognition challenges

INDUSTRY EXPERIENCE

Software Engineering Intern | DiGITs Program | Memorial Sloan Kettering Cancer Center (MSK)

Jun 2024 — Aug 2024

- Developed AI chatbot interface using React and TypeScript for patient portal serving **170,000+** users annually
- Streamlined LLM fine-tuning process for **50+** developer teams, improving AI integration across departments
- Led development of MSK SkinMonitor, **winning 1st place** in intern hackathon for AI-powered health tracking app
- Practiced Agile methodologies through daily stand-ups and two-week sprint cycles

PUBLICATIONS

- Navin Sriram Ravie, Andrew Jong, **Kayla McFarlane**, Jay Patrikar, Bijoy Sebastian, Sebastian Scherer. "Don't Fool Me Twice: Adapting to Failures in the Wild with Multi-Modal Reasoning." *IROS 2025 Workshop on The Art of Robustness: Surviving Failures in Robotics*, October 2025.

SKILLS

Programming: Python, C++, C, Java, JavaScript, TypeScript, MATLAB, SQL

Tools & Frameworks: ROS2, Isaac Gym/Sim, Gazebo, Unreal Engine, React, Git, Linux, Docker

Technical Areas: Robot Navigation, Computer Vision, Machine Learning, Safety-Critical Systems, Simulation